

Review

Advances in surrogate based modeling, feasibility analysis, and optimization: A review



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ABSTRACT

The idea of using a simpler surrogate to represent a complex phenomenon has gained increasing popularity over past three decades. Due to their ability to exploit the black-box nature of the problem and the attractive computational simplicity, surrogates have been studied by researchers in multiple scientific and engineering disciplines. Successful use of surrogates shall result in significant savings in terms of computational time and resources. However, with a wide variety of approaches available in the literature, the correct choice of surrogate is a difficult task. An important aspect of this choice is based on the type of problem at hand. This paper reviews recent advances in the area of surrogate models for problems in modeling, feasibility analysis, and optimization. Two of the frequently used surrogates, radial basis functions, and Kriging are tested on a variety of test problems. Finally, guidelines for the choice of appropriate surrogate model are discussed.

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1. Introduction

The problem discussed in the paper is assessing the performance of surrogates on the deterministic function $f: R^d \rightarrow R$; where the input vector is $X = (x_1, x_2, \dots, x_d)$, d is the number of dimensions of $X^L \leq X \leq X^U$ the problem, and there is a single output y . The input vector X has known lower and upper bounds

Additionally, some constraints $f_j \leq 0$, $j \in J$ where J is the set of all constraints, may be present. It is assumed that evaluation of the function as well as constraints is computationally expensive and the symbolic form of the function and that of one or more constraints is unknown. From this assumption, it follows that the analytical form of the derivatives is also unavailable. Surrogate modeling addresses this problem by obtaining a function $\hat{f}(X)$ that approximates the function f .

This problem occurs frequently in multiple engineering and scientific disciplines where complex computer simulations or physical experiments are used. In these cases, obtaining more data means additional experiments and thus it results in significant material or economic cost as well as highly non-trivial computational expense. As a result, it is difficult to obtain an analytical form of the objective function or that of the derivatives. Deriving this information from surrogate $\hat{f}(x)$ is relatively easier because its analytical form is known and it is cheaper to evaluate. Several applications of surrogates to address this type of problems can be found in the literature. For example, (Anthony et al., 1997), (Balabanov and Haftka, 1998), use polynomial, linear response surfaces in aircraft design. Artificial neural networks (ANN) is used for process modeling (Meert and Rijckaert, 1998), process control (Bloch and Denoeux, 2003), (Mujtaba et al., 2006), and for optimization (Fernandes, 2006), (Henao and Maravelias, 2011). Kriging is used for process flowsheet simulations (Palmer and Realf, 2002), design simulations (Yang et al., 2005), (Prebeg et al., 2014), pharmaceutical process simulations (Jia et al., 2009), and feasibility analysis (Rogers and Ierapetritou, 2015). Radial basis functions (RBF) is used for feasibility analysis (Wang and Ierapetritou, 2016) and parameter estimation (Müller et al., 2015). It can be observed from the $\hat{f}(x)$ applications listed above that there are multiple approaches proposed in literature to obtain a surrogate.

Several prior reviews discuss these approaches and related developments in the field of surrogate models. Surrogate models and their potential use in simulations is discussed by (Barton, 1992). They discuss polynomial response surface, spline interpolation, radial basis functions, regression models, and Kriging surrogates. With the focus on modeling and prediction for engineering design, (Simpson et al., 1997) review stationary sampling designs, polynomial response surface methods, Kriging and robust methods. (Jin et al., 2001) studied performance of polynomial regression, multivariate adaptive regression splines, radial basis functions, and Kriging surrogates under multiple criteria such as efficiency, robustness and model simplicity. Motivated from applications in

aerospace systems, (Queipo et al., 2005) discuss surrogate based optimization and sensitivity analysis, sampling strategies and surrogate model validation. (Barton and Meckesheimer, 2006) discuss surrogates for guiding optimization of simulations. In this context of guiding search towards optimum, they classify surrogates as local surrogates that are updated within an iterative framework and global surrogates that are fitted only once and the search proceeds using the same surrogate thereafter. For the purposes of design optimization, (Wang and Shan, 2007) provide an overview of surrogate models. Their focus is mainly on solving optimization problems such as global optimization, multi-objective optimization, and probabilistic design optimization. Motivated from computationally intensive aerospace designs, (Forrester and Keane, 2009) discuss details of surrogate modeling methodology focusing on sampling, surrogate model building, and validation. They discuss surrogates such as polynomial interpolation, RBF, Kriging and support vector regression and their advantages and disadvantages for achieving better prediction accuracy. (Razavi et al., 2012) investigate the potential of surrogate modeling techniques with a focus on the use of surrogates in water resources applications. They provide an excellent review on use of surrogates in water resources. (Nippgen et al., 2016) review surrogate modeling strategies from a broader point of view by classifying those as data-driven, projection-based and multi-fidelity surrogate modeling strategies. They focus on potential of using surrogates for applications in groundwater modeling. (Haftka et al., 2016) discuss in detail, several strategies for global optimization using surrogates, criteria for local and global searches from the point of view of parallelization. It is important to note that with respect to applications, the problems requiring surrogates can be classified in to three classes. The first class of problems is the most fundamental use of surrogates i.e. prediction and modeling. The second class of problems is commonly known as derivative-free optimization (DFO) where the objective function to be optimized is expensive and thus derivative information is unavailable. The third class of problems is feasibility analysis where the objective is also to satisfy design constraints. Prior reviews discuss applications of surrogate models for only one or two of these three classes. This review emphasizes that there is a significant difference between using surrogates for each of these three classes of problems and provides a comprehensive understanding of surrogate models for all three classes of problems mentioned. There has been a growing interest in model selection methodologies for regression models where the aim is to choose the best model from a given set of models. This problem has many practical uses in cases where the surrogate model does not generalize well on the test set, a phenomenon commonly known as overfitting or when there are too many input variables that might contain redundant information. In such cases, it is important to select most relevant variables in order to build simple yet effective surrogate models. Even though model selection is popular in the field of statistics for over 50 years, prior reviews in the con-

text of surrogate modeling do not address this issue. As avoiding overfitting an important aspect to consider while building surrogate models, an extensive review of model selection strategies is provided.

The rest of this paper is organized as follows. Section 2 discusses different types of surrogates and their underlying mathematical formulations. Sections 3 and 4 describe surrogates in relation to the DFO problems and feasibility analysis, respectively. Section 5 is devoted towards sampling which is an important component of building the right surrogate model. Section 6 describes approaches for validating surrogates. Section 7 describes some of the existing software implementations. In section 8, a detailed comparison of the performance of RBF and Kriging surrogates on a set of 47 test problems is provided whereas Section 9 provides a summary of the manuscript.

2. Surrogate models

In this section, frequently used approaches for obtaining the surrogate $\hat{f}(x)$ are discussed with a focus on the recent advances. The models that are designed to yield unbiased predictions at the sampled data are referred to as interpolation models, whereas models that are built by minimizing the error between given data and model prediction under a certain criterion are referred to as regression models. In this section, regression models such as linear regression, support vector regression are discussed followed by interpolation models such as RBF and Kriging. Finally, approaches utilizing more than one of these surrogates are discussed. With their ability to provide a quantitative measure of uncertainty in prediction, RBF and Kriging surrogates are the most popular choices for optimization and feasibility analysis. Therefore, special emphasis is given on these surrogates.

2.1. Linear regression

This is a commonly used approach where a surrogate is represented as a linear combination of the input variables as given by Eq. (1).

$$\hat{f}(x) = w_0 + \sum_{j=1}^d x_j w_j \quad (1)$$

where x is a vector of size d ; d is the number of variables; w is the vector of length $d+1$. To obtain the weight vector, sum of squared errors between the actual data and the surrogate predicted value is minimized. The unconstrained minimization problem can be formulated as given by Eq. (2).

$$\min_w \|Xw - y\|_2^2 \quad (2)$$

where X is a matrix of size n by $d+1$ where n is the number of sample points and all elements in the first column of X are 1 and columns 2 through $d+1$ correspond to the input vector; y is a vector of size n that represents function values at sample points. For the case of ordinary least squares, solution in analytical form is $w = (X^T X)^{-1} X^T y$. When one or more of independent variables are perfectly correlated, the matrix $X^T X$ becomes near singular. As a result, the coefficients w are not uniquely defined. This kind of rank deficiency can occur in high dimensional problems where the number of data points is less than the number of variables. This is usually addressed by reducing number of variables by screening or by utilizing regularization techniques. As the number of variables (d) in this problem increases, either inherently from the problem or from a combination of existing variables, this system is susceptible to produce high variance. Even though addition of extra variables leads to low bias on the data points used for building the model,

high variance makes it difficult to have better predictions on new data points. This phenomenon is known as overfitting. To avoid this issue, the effect of unnecessary variables is either removed using subset selection or suppressed using regularization. Subset selection and regularization strategies are explained below.

2.1.1. Subset selection

Subset selection refers to addressing the trade-off between prediction error and the regression model complexity by selecting a subset of variables. This step is followed by least squares regression for determining coefficients of the regression model. Several approaches for subset selection exist in the literature that are classified here as exhaustive search methods, heuristic methods, methods based on integer programming, methods relying on model fitness measures, Bayesian variable selection methods, and methods based on analyzing correlations between input variables and the output.

Exhaustive search methods try to exhaustively explore all possible subsets of features and select the subset with minimum prediction error. Advantage of exhaustive search is that a number of regression models are obtained with comparable prediction accuracy. Even though these methods guarantee the selection of best possible model, computational complexity of exhaustive search increases rapidly as the number of subsets increases. An implementation of this approach is the leaps and bounds algorithm (Furnival and Wilson, 1974).

Heuristic methods try to overcome this drawback by using greedy approaches such as forward-stepwise regression, backward-stepwise regression, and forward-stagewise regression. In forward-stepwise regression, variable selection starts from an empty set of variables and proceeds by sequentially adding a variable that improves the fit by largest magnitude. Improvement in the fit is usually measured by using the F-statistic. Using sum squared error, F-statistic quantifies the improvement achieved by addition of a new variable. Backward-stepwise regression is an opposite approach that starts from including all variables and sequentially removes variables that have least impact on the fit. Forward-stagewise regression is similar to forward-stepwise regression. However, in this case, only the coefficient of the newly added variable is adjusted keeping other coefficients constant.

Approaches that use integer programming for subset selection formulate the subset selection problem as an optimization problem. In these formulations, an error measure (EM) is minimized subject constraints that ensure subset selection. One way to impose such a constraint is by having an upper bound on the number of nonzero entries (Konno and Yamamoto, 2009a). In addition to limiting the number of nonzero entities, these formulations can be adapted to ensure statistical properties such as robustness, selectivity and general sparsity of the model (Bertsimas and King, 2016). These approaches, however, need a prespecified value for number of variables that might not be known a priori. Review of these approaches can be found in (Liu and Motoda, 2007). For a known number of variables to be selected, an example of problem formulation for these problems is given by Eqs. (3)–(5) (Konno and Yamamoto, 2009b).

$$\min \text{EM} \quad (3)$$

s.t.

$$\sum_{i=1}^d z_i = k \quad (4)$$

$$w_{lL} z_l \leq w_l \leq w_{lU} z_l, l = 1, \dots, d \quad (5)$$

$$z_l \in \{0, 1\}, l = 1, \dots, d \quad (6)$$

Table 1
Model fitness measures.

Model fitness measure	definition
AIC _c	$N \log \left(\frac{1}{N} \sum_{i=1}^N (y_i - X_i w)^2 \right) + 2p + \frac{2p(p+1)}{N-p-1}$
HQIC	$N \log \left(\frac{1}{N} \sum_{i=1}^N (y_i - X_i w)^2 \right) + 2p \log(\log(N))$
BIC _c	$\sum_{i=1}^N \frac{(y_i - X_i w)^2}{\hat{\sigma}^2} + p \log(N)$
RIC	$\sum_{i=1}^N \frac{(y_i - X_i w)^2}{\hat{\sigma}^2} + 2p \log(k)$
C _p	$\sum_{i=1}^N \frac{(y_i - X_i w)^2}{\hat{\sigma}^2} + 2p - N$

where, z_l is a binary variable for selection of variable l ; k is the number of subsets to be selected; for the coefficient w_l in the regression model, w_{lL} and w_{lU} represent the lower and upper bounds respectively. Eq. (4) limits the number of nonzero coefficients used in the model. Eq. (5) imposes bounds on w_l and forces w_l to be 0 when z_l is 0.

Methods utilizing model fitness measures tackle the issue of prespecifying number of selected variables by including a penalty for number of nonzero variables. This way these methods address the tradeoff between model complexity and prediction accuracy. Several fitness measures exist in literature. One such measure is mean absolute error (MAE). An algorithm to minimize MAE is proposed and used for finding subset of variables (Park and Klabjan, 2013). Other such measures include Mallows's Cp (Mallows, 1973), Akaike information criterion (AIC) (Akaike, 1974), Bayesian information criterion (BIC) (Schwarz, 1978), the Hannan-Quinn information criterion (HQIC) (Hannan and Quinn, 1979), the risk inflation criterion (RIC) (Foster and George, 1994), and mean squared error (MSE). These measures are shown in Table 1. These fitness measures can be used as EM in Eq. (3) to form a mixed integer quadratic program (Wilson and Sahinidis, 2017), (Cozad et al., 2014). AIC is based on the idea of minimizing discrepancy between the original distribution of the data and the distribution given by linear regression model. A well-known discrepancy measure called Kullback-Leibler divergence is used. Other discrepancy measures include Kolmogorov-smirnov and Hellinger discrepancy (Linhart and Zucchini, 1986). AIC_c represents correction term added to AIC for finite sample sizes (Hurvich and Tsai, 1989). C_p simply tries to minimize prediction error where mean squared error is the error measure. BIC_c seeks to maximize approximate posterior probability. These metrics are tabulated in (Table 1) where, $p < k$ is the number of coefficients, N is the number of sampled points, $\hat{\sigma}^2$ is an estimate of the error variance.

Bayesian approach models the uncertainty over unknowns in a surrogate model using probability theory assuming those as random variables. The probability distribution that represents this uncertainty before obtaining samples is referred to as prior distribution and that after obtaining samples is referred to as posterior distribution. Suppose that there are M models under consideration where i^{th} model is θ_m represented by $\hat{f}_i$ and unknowns corresponding to each model are represented as

The aim is to select a model with highest posterior probability, which is given by Eq. (7).

$$\Pr(\hat{f}_m | X) = \frac{\Pr(X | \hat{f}_m) \Pr(\hat{f}_m)}{\sum_{k \in M} \Pr(X | \hat{f}_k) \Pr(\hat{f}_k)} \quad (7)$$

where,

$$\Pr(X | \hat{f}_k) = \int \Pr(X | \theta_k, \hat{f}_k) \Pr(\theta_k | \hat{f}_k) d\theta_k$$

and X is sampled data set, θ_k represents unknowns in surrogate model $\hat{f}_k$. Bayesian variable selection problem is usually considered as a special case of the model selection problem where each model consists of a subset of variables. It should be noted that for comparison between two candidate models, the denominator on the right-hand side of Eq. (7) is the same and therefore, only numerator is usually considered for comparison. Finding the model with highest posterior probability is the fundamental motivation behind BIC.

Another class of subset selection approaches is the one that relies on learning the correlation between input variables and the output. One such method is sure independence screening (SIS). Sure independence screening (SIS) relies on learning ranks of input variables according to their marginal correlation with output y . After standardizing columns of matrix X , where each column corresponds to each input variable, a vector $X^T y$ is obtained which directly signifies marginal correlations of input variables with the output. With this method, input variables having least impact on the output can be filtered out. Another famous approach involving a similar strategy of assessing impact of an input variable by monitoring correlation with output is least angles regression (Efron et al., 2004). In this approach, coefficients are added in a similar fashion to forward-stepwise regression. However, instead of obtaining a least squares solution, correlation with the output is monitored and new variables are added sequentially.

Finally, subset selection is extremely important especially when the number of input variables is much higher than the size of available data set. Few of the approaches to address this class of problems include Dantzig selector (Candes and Tao, 2007), adaptive lasso and sure independence screening (Fan and Lv, 2008). For high dimensional problems, (Cadima et al., 2004) review heuristic algorithms for subset selection. An extension of BIC for high dimensional problems known as extended BIC is proposed (Chen, 2014).

2.1.2. Regularization

Subset selection methods lead to a discrete decision of either accepting or discarding a certain variable. This leads to high variance in prediction and does not reduce the prediction error of the regression model. Regularization leads to a continuous reduction of the regression model coefficients.

Regularization penalizes magnitude of regression coefficients w to modify the problem given in Eq. (2) to the form given in Eq. (8) (Hastie et al., 2009).

$$\min_w \|Xw - y\|_2^2 + C\|w\|_q \quad (8)$$

where,

$$\|w\|_q = \left(\sum_{i=1}^d w_i^q \right)^{1/q} \quad (9)$$

and C is the parameter that decides magnitude of regularization. Eq. (9) is the expression of q^{th} norm. Value of q has a significant effect on properties of the regression model. Values 1 and 2 represent two commonly used variants of regularization known as lasso and ridge regression respectively (Seber and Lee, 2003). In contrast to ridge regression, lasso has the ability to set regression coefficients exactly to 0. Values of q between 1 and 2 describe a mix between properties of lasso and ridge regression. Another approach to obtain a similar

mix is known as elastic-net regression where a linear combination of lasso and ridge regression penalty terms is used.

$$\min_w \|Xw - y\|_2^2 + C \sum_{j=1}^d (\alpha w^2 + (1 - \alpha) |w_j|) \quad (10)$$

where, α is a tuning parameter. Other extensions of Lasso include adaptive lasso (Zou, 2006) where the penalty term is a weighted summation where the weight depends on magnitude of the coefficient itself. Another for reducing the absolute value of regression coefficients uses a non-negative garrotte estimator (Breiman, 1995). This estimator is obtained by scaling coefficients of least squared regression. A penalty is associated with the scaling parameters and the problem is to find these scaling parameters. In this case, a closed form expression for these parameters is available as a function of coefficients obtained using ordinary least squares.

2.2. Support vector regression

The Support Vector Regression (SVR) surrogates are represented as the weighted sum of basis functions added to a constant term. A general form of SVR surrogate is given in Eq. (11).

$$\hat{f}(X) = \mu + \sum_{i=1}^n w^i \psi(X, X^i) \quad (11)$$

Assuming a simple basis function $\psi(\cdot) = X$, the surrogate can be written as per Eq. (12).

$$\hat{f}(x) = \mu + w^T X \quad (12)$$

This form of the surrogate is similar to that of RBF as well as Kriging. However, the way to calculate unknown parameters for this surrogate differs significantly from that of RBF and Kriging surrogates. The unknown parameters μ and w in the model are obtained by formulating a mathematical optimization problem given by Eqs. (13)–(16).

$$\min \frac{1}{2} |w|^2 + C \frac{1}{n} \sum_{i=1}^n (\xi^{+(i)} + \xi^{-(i)}) \quad (13)$$

s.t.

$$w \cdot x^i + \mu - y^i \leq \varepsilon + \xi^{-(i)} \quad (14)$$

$$y^i - w \cdot x^i - \mu \leq \varepsilon + \xi^{+(i)} \quad (15)$$

$$\xi^{+(i)}, \xi^{-(i)} \geq 0 \quad (16)$$

Eqs. (14) and (15) allow the sample points to lie within $\pm \varepsilon$ deviation from the value at sampled points without affecting the surrogate model. This band of allowed deviation is referred to as ε insensitive tube. Slack variables $\xi^{+(i)}$ and $\xi^{-(i)}$ ensure feasibility of the problem by allowing outliers that do not fall within ε insensitive tube. Trade-off between model complexity and fit is achieved by penalizing outliers by a pre-defined constant $C \geq 0$. Combined contribution of the model complexity and the penalty for outliers (Eq. (13)) is minimized.

The above-mentioned formulation is obtained under the assumption of a linear basis function. Using a different basis function might require determining additional hyper-parameters associated with that specific basis function. Details and mathematical derivations related to SVR can be found in (Smola and Schölkopf, 2004).

Finally, SVR is shown to achieve comparable accuracy with that of other surrogates (Clarke et al., 2005). SVR models are accurate as well as fast in prediction; however, the time required to build this model is high because finding the unknown parameters requires

Table 2

Commonly used basis functions.

Type	Function $\phi(\cdot)$
Linear	$\phi(r) = r$
Cubic	$\phi(r) = r^3$
Thin plate spline	$\phi(r) = r^2 \log(r)$
Multi-quadratic	$\phi(r) = \sqrt{r^2 + \gamma^2}$
Gaussian	$\phi(r) = e^{-\gamma r^2}$

solving a quadratic programming problem. This added complexity hinders the popularity of SVR (Forrester and Keane, 2009).

2.3. Radial basis functions

Given n distinct sample points, RBF surrogates can be represented as given in Eq. (17).

$$\hat{f}(x) = \sum_{i=1}^n \lambda_i \phi(\|x - x_i\|_2) + p(x) \quad (17)$$

where $\lambda_1, \dots, \lambda_n \in R$ are the weights to be determined; $\|\cdot\|$ is the Euclidean norm; $\phi(\cdot)$ is the basis function. There are several options for choosing the basis function $\phi(\cdot)$ as shown in Table 2.

In the case of multi-quadratic and Gaussian basis functions, $r \geq 0$, and γ is a positive constant. There is no solid conclusion in literature that decisively concludes one of these basis functions is better than others. However, use of cubic basis function with linear tail has been found to be successful (Björkman and Holmström, 2000). It can be represented by Eq. (18).

$$\hat{f}(x) = \sum_{i=1}^n \lambda_i \phi(\|x - x_i\|_2) + a^T x + b \quad (18)$$

The weights λ , a and b in Eq. (18) can be determined uniquely by solving the system of equations given by Eq. (19).

$$\begin{pmatrix} \phi & P \\ p^T & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ c \end{pmatrix} = \begin{pmatrix} F \\ 0 \end{pmatrix} \quad (19)$$

where ϕ is an n by n matrix with $\phi_{ij} = \phi(\|x - x_i\|_2)$;

$$p = \begin{pmatrix} x_1^T & 1 \\ x_2^T & 1 \\ \vdots & \vdots \\ x_n^T & 1 \end{pmatrix}; \quad \lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}; \quad c = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_d \\ a \end{pmatrix}; \quad F = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix}$$

An extension of RBF for the purposes of global optimization using a function called the bumpiness function (described in Section 5.2) is proposed (Gutmann, 2001). Several variations of this approach are discussed in Section 5 (Björkman and Holmström, 2000), (Regis and Shoemaker, 2007).

2.4. Kriging

Kriging is based on the idea that a surrogate can be represented as a realization of a stochastic process. This idea was first proposed in the field of geostatistics by (Krige, 1951) and (Matheron, 1963). It gained popularity after being used for design and analysis of computer experiments by (Sacks et al., 1989). Kriging is also known as Gaussian process regression in the field of machine learning (Rasmussen and Williams, 2006a).

A Kriging surrogate in its most general form can be formulated as:

$$\hat{f}(x) = \sum_{i=1}^m \beta_i f_i(x) + \varepsilon(x) \quad (20)$$

where $f_i(x)$ are m known independent basis functions that define the trend of mean prediction at location x ; β_i are unknown parameters; $\varepsilon(x)$ is a random error at location x that is normally distributed with zero mean. Kriging predictor is written as

$$\hat{f}(x) = f(x)^T \beta^* + r(x)^T \gamma^* \quad (21)$$

where, $f(x) = [f_1(x), \dots, f_m(x)]^T$; β^* is the vector of generalized least-square estimates of $\beta = [\beta_1, \dots, \beta_m]^T$; $r(x)$ is the correlation vector of size $n \times 1$ between $\varepsilon(x)$ and $\varepsilon(x_i)$. β^* and γ^* are given as:

$$\beta^* = (F^T R^{-1} F)^{-1} F^T R^{-1} y \quad (22)$$

$$\gamma^* = R^{-1} (y - F \beta^*) \quad (23)$$

where, R is the covariance matrix of size $n \times n$ where (i, j) element is the correlation between $\varepsilon(x_i)$ and $\varepsilon(x_j)$; $F = [f(x^{(1)}), \dots, f(x^{(n)})]^T$ is $n \times m$ matrix; y are observations at available data.

Estimate of the variance of prediction is given by:

$$s^2(x) = \hat{\sigma}^2 [1 - r^T R^{-1} r] \quad (24)$$

where, $\hat{\sigma}^2 = \frac{1}{n} (y - F \beta^*)^T R^{-1} (y - F \beta^*)$.

The correlation model used for obtaining R and r depends on a set of unknowns also known as hyper-parameters. These hyper-parameters are estimated using maximum likelihood (ML). For convenience, the log ML estimate (Eq. (25)) is often used:

$$\begin{aligned} \log ML(\theta) = & -\frac{1}{2} [\ln(2\pi\sigma^2) + \ln \det(R(\theta)) \\ & + (y - F\beta^*)^T R(\theta)^{-1} (y - F\beta^*) / \sigma^2] \end{aligned} \quad (25)$$

Having a random error term allows Kriging surrogates to provide an estimate of uncertainty in addition to the predicted value at a specific location.

2.4.1. Variants of Kriging

Kriging surrogate shown in Eq. (20) consists of regression component given by $\sum_{i=1}^m \beta_i f_i(x)$ and correlation component implied by ε . Several choices for both of these components are proposed in literature combinations of which lead to multiple variants of Kriging.

2.4.1.1. Correlation models. The random variables $\varepsilon(x)$ in a Kriging surrogate are assumed to be correlated according to a correlation model. For a deterministic and continuous function, if two samples are close to each other, their predicted values are close. As a result, the correlation between random variables is high. Correlation models consider the effect that the correlation decreases as the distance between two distinct samples increases. Commonly used correlation models are depicted in Table 3 where m_j is the distance between two points; θ_j and p_j are hyper-parameters; d is the number of dimensions of the original problem. In case of Matern correlation model, Γ is the Gamma function, K_{ν_j} is the modified Bessel function of order ν_j . The parameter $\nu_j > 0$ provides control over the differentiability of correlation model with respect to input variable x_j and therefore that of the Kriging predictor. (Chen et al., 2013) compare some of these correlation models and their results show that the squared exponential correlation performs worse than the exponential correlation. However, it is important to note that

Table 3

Commonly used correlation models in Kriging surrogates.

Name	Correlation model
Exponential	$\exp(-\sum_{j=1}^d \theta_j m_j ^{p_j}), 0 < p_j < 2$
Squared exponential	$\exp(-\sum_{j=1}^d \theta_j m_j ^2)$
Linear	$\max \left\{ 0, 1 - \sum_{j=1}^d \theta_j m_j \right\}$
Spherical	$1 - 1.5\xi_j + 0.5\xi_j^3, \xi_j = \min \left\{ 1, \sum_{j=1}^d \theta_j m_j \right\}$
Matern	$\prod_{j=1}^d \frac{1}{(\nu_j)^{2\nu_j-1}} (\theta_j m_j)^{\nu_j} K_{\nu_j}(\theta_j m_j)$

the generalized exponential correlation model has a higher number of hyper-parameters ($2d$) as opposed to d in case of squared exponential correlation. They also suggest choosing Matern correlation model (Table 3) as an alternative to exponential correlation model. Differentiability of this correlation model can be controlled by choosing an appropriate value of ν . For example, $\nu_j = 1 + \frac{1}{2}$ or $\nu_j = 2 + \frac{1}{2}$ make sure that there are 1 or 2 derivatives of the correlation model respectively.

2.4.1.2. Regression models. Based on the choice of the mean prediction model $f(x)^T \beta$ (given in Eq. (20)) there are several variants of Kriging such as simple Kriging, ordinary Kriging, and universal Kriging (also known as Kriging with a trend). Simple Kriging assumes the term $f(x)^T \beta$ to be a known constant, ordinary Kriging assumes it to be an unknown constant, and universal Kriging assumes $f(x)$ to be any other prespecified function of x . In universal Kriging, usually, $f(x)$ takes form of a lower order polynomial regression. However, specifying a trend or a value for the mean when the underlying function is unknown may lead to inaccuracy in prediction. To avoid this problem blind Kriging is used (Joseph et al., 2008). In blind Kriging, the unknown trend is identified using a Bayesian variable selection technique. From a given set of candidate models, Bayesian approach tries to select models that have maximum posterior probability (Section 2.1.1). Several other approaches for variable selection exist in the literature for blind Kriging. For example, (Huang and Chen, 2007) propose a metric known as generalized degrees of freedom which is an estimator of mean squared error. Variable selection is done by trying to minimize this estimator.

There are a few strategies developed based on penalized likelihood function for variable selection in Kriging where the idea of adding a penalty term in regularization (discussed in Section 2.1.1) is implemented in the context of likelihood functions (Fan and Li, 2001). Unlike penalized least squares approaches discussed in Section 2.1.2, algorithms involving penalized likelihood functions involve operations with covariance matrix which is of a size of the order of size of sampled data set (discussed in Section 2.4.3). As this problem occurs frequently in building Kriging surrogates, efficient optimization algorithms are developed specifically for this problem (Zou and Li, 2008; Chu et al., 2011; Furrer et al., 2006; Kaufman et al., 2008).

By having different combinations of mean prediction terms $f(x)$ and correlation models used for random error $\varepsilon(x)$, multiple Kriging models can be obtained. One such comparison between Kriging models was made by (Chen et al., 2013). For regression terms, their results reveal that adding complex regression terms to Kriging might not be of advantage over ordinary Kriging in terms of predic-

tion accuracy. Moreover, adding these complex terms might result in multimodal ML function, thus adding an extra computational expense (Section 2.4.3).

2.4.2. Nugget effect

Kriging by its fundamental problem formulation is an exact interpolation technique. This means that Kriging surrogate predicted value matches exactly with the underlying black-box function at the sample points used to build the Kriging model. This nature of Kriging might lead to highly oscillating behavior of the prediction. To suppress this, Kriging regression is an approach that attempts to add a regression component to Kriging.

In this approach, the covariance matrix is augmented by a term known as the nugget. The effect of this added term on Kriging surrogates is known as nugget effect. Mathematically, the correlation matrix obtained after adding the nugget term ε is shown in Eq. (26).

$$R^{\text{mod}} = R + \varepsilon I \quad (26)$$

Because of this modification, as distance between two points approaches zero, the correlation no longer equals 1. A singular or ill-conditioned covariance matrix occurs when two of the sample locations are very close to each other or hyper-parameters in the covariance model are near zero (Ranjan et al., 2011). Incorporating nugget effect in such cases helps in maintaining conditioning of covariance matrix. The remainder of the procedure to obtain Kriging predictor remains the same as before.

2.4.3. Computational aspects of kriging

A few key computational aspects of Kriging need to be understood before choosing Kriging surrogate for the problem at hand. First, obtaining a Kriging surrogate involves inversion of a covariance matrix. The size of this matrix depends on the number of samples and thus its inversion may become computationally demanding as the number of samples grows. Second, to obtain hyper-parameters of the correlation model, Kriging maximum likelihood (ML) estimator is optimized. This ML estimator is highly non-convex and has a strong dependence on the inverse of the covariance matrix. The non-convex nature of this function demands multiple evaluations to search for global optima. Couple of approaches are proposed to tackle this problem with likelihood maximization can be found in (Toal et al., 2011; Toal et al., 2009). From the equations, one can observe that getting stuck at a local optimum affects Kriging surrogate prediction as well as quantified uncertainty at unsampled locations. However, with simple covariance functions, experience shows that getting stuck at local optimum is not a serious problem and often there is no point in finding the minimizer with great accuracy (Rasmussen and Williams, 2006b), (Lophaven et al., 2002a,b).

The non-convex and computationally intensive nature of ML estimator becomes a bigger problem as the dimensionality of the problem increases. It can be observed from Table 3, that irrespective of the correlation model chosen, the number of hyper-parameters, depends on the dimensionality of the problem. To reduce the number of hyper-parameters, (Bouhrel et al., 2015) use partial least squares. This way they address problems up to 100 dimensions more efficiently than other existing approaches. Another way to optimize hyper-parameters is to use cross validation instead of maximum likelihood. Use of cross-validation is found to be more robust with respect to correlation model misspecification compared to using maximum likelihood. However, the variance obtained by Kriging surrogates employing cross validation is larger (Bachoc, 2013).

For the problem with large number of data points, there are several successful applications of Kriging in the literature. One way to do this is by representing covariance matrix in terms of small matrices of size r , where r is the number of basis functions used

(Cressie and Johannesson, 2008). Similar approach of reducing size of covariance matrix from number of sample points (n) to a smaller number r is used by (Nychka et al., 2015), and (Banerjee et al., 2008). Another approach is using covariance tapering, where a sparse covariance matrix is obtained by setting majority of the insignificant elements to zero. Sparse matrix inversion techniques are then used to achieve attractive computational complexity (Furrer et al., 2006). Another way is to choose only a subset of data for building Kriging model (Liang et al., 2013). There exists a large amount of literature for using Kriging on large datasets by combination of the above-mentioned approaches (Snelson and Ghahramani, 2007), (Sang and Huang, 2012) or by other frameworks (Tajbakhsh et al., 2014).

Finally, even though inversion of covariance matrix is a computationally intensive task, positive definiteness of the covariance matrix helps current software implementations reduce the computational complexity by a significant factor. For obtaining maximum likelihood within a limited number of function evaluations, some software implementations make use of DFO algorithms.

2.5. Mixture of surrogates

Realizing the fact that no single type of surrogates outperforms all other types for all types of problems, choosing the best type of surrogate for the problem at hand is a challenging task. It is not always possible to try multiple choices of surrogate models and choose the surrogate model that shows the best performance. This motivates approaches utilizing a combination of surrogates. In general, prediction using a mixture of surrogates can be given by Eq. (27).

$$\hat{f}(x) = \sum_{i=1}^n w_i(x) \hat{f}_i(x) \quad (27)$$

where, $w_i(x)$ is the weight associated with the i^{th} surrogate at design point x . Finally, summation of weights is set to one $\sum_{i=1}^N w_i = 1$

1. This implies that if all surrogate predictions $\hat{f}_i(x)$ are equal, the weighted mixture will predict the same value.

Different approaches to determine the weights w_i are used in the literature. For example, (Zerpa et al., 2005) use a mixture of surrogates to optimize alkaline-surfactant-polymer flooding processes. They use a weighted combination of Kriging, RBF and polynomial regression where weights are determined based on the variance of individual surrogates. Weights can also be determined using a global cross validation metric called PRESS (discussed in Section 6) (Goel et al., 2007). Another approach for identifying weights is by weighing the surrogates with the help of an error metric proposed by (Juliane Müller and Piché, 2011). They assign probability to surrogates with the help of an error metric. These probability assignments are then used to determine weights. A variant of efficient global optimization utilizes mixture of surrogates (Felipe A C Viana et al., 2013). They propose multiple surrogate efficient global optimization approach that is able add multiple candidate points for global optimization in a single iteration. Use of multiple surrogates, in general, provides a flexibility to emphasize more on good surrogates and put less emphasis on bad surrogates as per the need.

3. Derivative-free optimization and surrogates

The optimization problems for which function derivative information is not symbolically or numerically available are classified as DFO problems. There are two sub-categories in algorithms addressing DFO problems, one is local search (referred to as local DFO)

algorithms and the other is global search algorithms (referred to as global DFO). Local search algorithms are effective in refining the solution or reaching a local optimum from an initial guess. Global search algorithms, on the other hand, have a component that allows escaping from a local minimum. For the purposes of this paper, it is convenient to classify DFO algorithms as algorithms that do not use surrogate models and model-based algorithms. A major class of local DFO algorithms that do not rely on surrogate models is direct search algorithms. Direct search algorithms sequentially examine candidate points generated by a certain strategy sometimes recognizing geometric patterns. Well known examples of the direct search are Hooke and Jeeves's algorithm (Hooke and Jeeves, 1960) and simplex method (Nelder and Mead, 1965). Model-based approaches, as the name suggests, rely on surrogate models to guide the search. For the case of global DFO algorithms, the majority of the algorithms that do not use surrogate models use an approach such as partitioning of the feasible space or a stochastic approach. An example of partitioning algorithm is DDividing RECTangles (DIRECT) algorithm (Jones et al., 1993). Examples of stochastic algorithms include several approaches such as simulated annealing or genetic algorithms. For details on advances in DFO and an extensive comparative study on box-bounded problems, the readers are referred to (Rios and Sahinidis, 2013).

As model-based search algorithms have been shown to display superior performance compared to these algorithms, it is important to discuss the role of surrogate models in the context of DFO. A major class of model-based local DFO methods known as trust-region methods is discussed followed by model-based global DFO methods.

3.1. Model-based local DFO

Trust-region methods are local search methods that rely on a surrogate model in a neighborhood of a given sample location. This neighborhood is called as trust region and the model is presumed to be accurate within trust region. The size of the trust region is defined with the help of radius which is adjusted based on a measure of the accuracy of the surrogate. The sufficiently small value of trust-region radius usually indicates termination. Because of the general nature of trust-region framework, several surrogates have been used in the literature to achieve local approximation. For example, (Powell, 1994) use linear interpolation models to approximate objective and constraint functions in the algorithm COBYLA (Constrained Optimization BY Linear Approximation). Linear Interpolating surrogates are easy to construct but these surrogates face difficulty in capturing curvature of the original problem. Use of quadratic models of the form given in Eq. (28) is proposed (Conn et al., 1997).

$$\hat{f}(x_k + s) = f(x_k) + s^T g_k + \frac{1}{2} s^T H_k s \quad (28)$$

where, k corresponds to iteration k , x_k is the current iterate, $g_k \in \mathbb{R}^d$, H_k is a matrix of size d by d . Uniquely determining g_k and H_k requires $\frac{(d+1)(d+2)}{2}$ sample points. This number becomes significantly high as the number of dimensions increase. For example, for a 30 dimensional problem, this number becomes nearly 500. To avoid this high sampling requirement, (Powell, 2006) proposed underdetermined quadratic interpolation models. These models are proven to attain stationary local optimum and thus are called locally convergent. Further, (Ouvray and Bierlaire, 2009) use RBF interpolation models with cubic basis functions and a linear tail. With modifications to the set of points used for building RBF models, (Wild et al., 2008) proposed Optimization by RBF Interpolation in Trust-regions (ORBIT) algorithm. This algorithm was later extended to handle constrained optimization problems (Regis and Wild, 2015). RBF based trust region algorithms are proven to be globally con-

vergent (Wild and Shoemaker, 2013). A similar strategy was used recently where Kriging based efficient global optimization (EGO) where Kriging surrogate was used inside trust-region framework (Regis, 2016).

3.2. Model-based global DFO

One of the reasons surrogates are promising in the context of global DFO is the progress made in the area of global optimization algorithms in the past decade (Tawarmalani and Sahinidis, 2005), (Misener and Floudas, 2014). With the help of these algorithms, non-convex surrogates can be optimized and used to guide the search. For global DFO, a surrogate model is generated over the entire feasible space or multiple parts of the feasible space. For example, in the algorithm EGO (Jones et al., 1998), Kriging surrogate is built over the entire feasible space. With the help of expected improvement (EI) function (discussed in Section 5.1), the surrogate is updated. In this case, a balance between local search and global search is achieved by maximizing the EI function. A similar balance is achieved for RBF surrogates using a measure called bumpiness function (Eq. (33)) (Gutmann, 2001). Minimizing the bumpiness function for a given target value can be used to focus on global and local search. This property of the bumpiness function is exploited for global DFO (Holmström et al., 2008). EI function, as well as bumpiness function, are discussed in more detail in Section 5. Another approach using RBF surrogate relies on optimization and sequential updating of RBF surrogate over the feasible space (Regis and Shoemaker, 2005). Finally, global search is also achieved by conducting local search starting from multiple starting points obtained using a certain strategy. A complete restart strategy is proposed that suggests starting from a new sample design if algorithm gets stuck in a local minimum (Regis and Shoemaker, 2007). A more recent example of successful use of surrogates to address constrained global DFO problems can be found in (Boukouvala and Floudas, 2016). They developed a framework for constrained grey box optimization named Algorithms for Global Optimization of coNstrained grey-box compUTational problems (ARGONAUT) that was shown to address a difficult class of problems successfully. In this framework, the surrogate is chosen from a set containing linear, general quadratic, sigmoidal, RBF and Kriging models based on the accuracy of prediction. Details of this framework are discussed in Section 7.

4. Feasibility analysis and surrogates

The ability of a process to satisfy all relevant constraints is referred to as feasibility. Feasibility analysis relates to identifying conditions under which the process is feasible. The problem of feasibility analysis arises because of several constraints on operation such as product demand, environmental conditions, the safety of operation, and material properties to name a few. The same problem arises in designing a product or a new material where the design is restricted by factors such as target quality. To conduct a systematic study of profit maximization or to compare multiple design alternatives, a precise estimation of feasible operation regions is crucial. Therefore, understanding the problem of feasibility analysis is extremely important.

Mathematically, feasibility is quantified with the help of a measure known as the feasibility function $\psi(d, \theta)$ as given in Eq. (29). A positive value of the feasibility function implies that the design is infeasible.

$$\psi(d, \theta) = \min_z \max_{j \in J} \{f_j(d, z, \theta)\} \quad (29)$$

where d and z represent design variables and control variables respectively, bounds on z are written as $z \in Z = \{z : z^L \leq z \leq z^U\}$;

θ represents uncertain parameters $\theta \in T = \{\theta : \theta^L \leq \theta \leq \theta^U\}$; $f_j(d, z, \theta)$ represents constraints. The problem is to check if all constraints f_j can be satisfied for a given design d by adjusting the control variables z . Thus, $\psi(d, \theta) > 0$ implies one or more constraints are violated and $\psi(d, \theta) = 0$ implies boundary of the feasible region.

Uncertainty in process input parameters θ has a great impact on feasibility of the process design. The ability of a process to remain feasible when subject to nominal deviations of uncertain parameters is referred to as process flexibility. Process flexibility is quantified by solving the flexibility test problem which checks if feasibility function $\psi(d, \theta)$ is non-positive over the entire range of uncertain parameters θ . The flexibility test problem is usually represented as a max-min-max problem as represented in Eq. (30)

$$\chi(d) = \max_{\theta} \min_z \max_{j \in J} \{f_j(d, z, \theta)\} \quad (30)$$

Several approaches utilizing the concept of feasibility function and flexibility are available in the literature. (Swaney and Grossmann, 1985) quantify feasible region by means of the largest hyperrectangle that inscribes the feasible region. Other approaches for feasibility analysis given the closed form of the process include (Straub and Grossmann, 1993), (Floudas and Gümes, 2001), (Goyal and Ierapetritou, 2002). A good review of these approaches can be found in (Grossmann et al., 2014). In these approaches, analytical form of the feasibility function is assumed to be known. However, simulation models for which the feasibility needs to be assessed often require substantial computational efforts. For this reason, the closed form of these models is not available or cannot be utilized. This motivates approaches involving black-box feasibility analysis. The black-box nature of the problem and computational complexity of the underlying simulations make surrogate-based methods an intuitive choice for the black-box feasibility analysis.

The key idea of using surrogate based feasibility analysis is building a surrogate to represent feasibility function given a set of input parameters and black-box simulation outputs for the feasibility function. Two key factors for this class of problems are the choice of surrogate model and sampling strategy. Several approaches exist in the literature to address both of these factors. For example, (Banerjee et al., 2010) use high dimensional model representation (HDMR), (Boukouvala and Ierapetritou, 2012) use Kriging surrogates, (Wang and Ierapetritou, 2016) use RBF to represent feasibility function over the entire domain. In addition to these, for mixed-integer programming frameworks, (Zhang et al., 2016) propose convex region surrogate (CRS) for representing a nonlinear and nonconvex feasible region by a combination of convex regions. They approximate cost function for each region by a linear approximation. (Adi et al., 2016) use a random line search for detecting boundary points of the feasible region.

As described in Section 5, quality of surrogates has a strong dependence on the required quantity and quality of sampling set. Increasing sample size may lead to a better prediction but it will result in increased sampling cost. For feasibility analysis problems, sampling requirement is higher than that of single objective prediction due to the presence of constraints. To control the sampling cost, approaches employing adaptive sampling are used. Kriging surrogates and a modified version of EI function given in Eq. (31) for adaptive sampling is proposed (Boukouvala and Ierapetritou, 2014).

$$\max_x \text{EI}_{\text{feas}}(x) = s\phi\left(-\frac{y}{x}\right) = s\left(\frac{1}{\sqrt{2}}e^{-0.5\left(\frac{y^2}{x^2}\right)}\right) \quad (31)$$

where $\text{EI}_{\text{feas}}(x)$ is the modified EI function value at x ; y is the surrogate model predictor; s is the standard error of the predictor; $\phi(\cdot)$ is the normal probability distribution function.

There is a difference between a search for global optimization and that for feasibility analysis. For feasibility analysis, the problem is to find a surface defining the boundary of the feasible space within the box bounded design space as opposed to finding a single optimum in global optimization (Boukouvala and Ierapetritou, 2012). To address this problem, (Boukouvala and Ierapetritou, 2012) use an approach that utilizes Kriging variance to ensure exploration. To guide the search towards better defining the boundary of feasible space, they use the product of feasibility function values of nearby samples. Samples on the same side of feasible boundary result in a positive product. More recently, (Wang and Ierapetritou, 2016) used an adaptive sampling strategy based on RBF surrogates. They used bumpiness measure (explained in Section 5.2) to obtain prediction error. Substituting this prediction error in EI_{feas} function and maximizing EI_{feas} with respect to x , they chose new sample points for evaluation. Their results show that accuracy obtained from both Kriging and RBF is comparable.

5. Sampling

The process of generating data points to be able to build surrogates is referred to as sampling. The performance of surrogate models depends strongly on the quality as well as the number of samples. However, as generating data demands evaluation of the true function, sampling contributes towards significant computational cost. To maintain the quality of surrogates without incurring excessive sampling cost, studying sampling strategies is of immense importance.

Sampling strategies are broadly classified as adaptive sampling and stationary sampling. Stationary sampling consists of methods that rely on geometry or pattern such as grid sampling, full and half factorial designs, methods that were derived from the design of experiments literature such as orthogonal sampling, full and half factorial designs, Box-Behnken design. Some of the widely used stationary sampling strategies are Latin Hypercube Sampling (LHS) (McKay et al., 1979), Sobol (1967) and Halton sampling. LHS is a stratified sampling strategy where a sample is drawn from each stratum once. To provide better space-filling properties, LHS is done subject to projection filters. Sobol and Halton sampling are quasi-random strategies where samples are drawn from Sobol and Halton low-discrepancy sequences respectively.

In adaptive sampling, starting from a limited number of samples that are generally obtained from stationary sampling, new sample locations are decided sequentially. This strategy aims to minimize sampling requirement by obtaining more samples that benefit the quality of the surrogate. Most of the new adaptive sampling strategies rely on some criteria to tackle the trade-off between exploring the most unexplored region (exploration) and refining the region near existing samples for better understanding (exploitation). This approach is most common in the context of global optimization where exploration is required to escape local optima and exploitation is required to improve available optimum. For Kriging surrogates, a popular approach is making the use of EI function. For RBF surrogates, a similar quantitative measure is obtained using a function known as bumpiness function. Other approaches employing adaptive sampling make use of different strategies to address this trade-off. In general, these methods have been shown to achieve better accuracy with fewer samples (Provost et al., 1999).

5.1. Expected improvement function

One commonly used approach to handling exploration and exploitation is by using EI function (Eq. (32)) as used by (Donald

R Jones et al., 1998).

$$EI(x) = (f_{\min} - \hat{f}(x)) \phi \left(\frac{f_{\min} - \hat{f}(x)}{s} \right) + s \phi \left(\frac{f_{\min} - \hat{f}(x)}{s} \right) \quad (32)$$

where, $\phi(\cdot)$ represents the standard normal density function; $\phi(\cdot)$ represents the probability distribution function; $\hat{f}$ is the surrogate model predictor; $f_{\min}$ is the current minimum function value and s is the standard deviation. $EI(x)$ represents the expected improvement at sample location x . The function increases with decreasing $\hat{f}(x)$ that corresponds to the predicted value and increasing standard deviation s . Achieving low $\hat{f}(x)$ and high s correspond to exploration and exploitation, respectively. As both contribute positively towards EI function, the trade-off between exploration and exploitation is addressed by maximizing the EI function. The EI function exhibits multiple local optima that might cause numerical problems.

5.2. Bumpiness function

A similar approach of having a single function to balance exploration and exploitation was proposed by (Gutmann, 2001) for RBF surrogates. This relies on the fact that the RBF surrogate that is obtained by solving the system of equations given by Eq. (19) is the one that minimizes bumpiness. A quantitative measure of bumpiness is given by bumpiness function (Eq. (33)).

$$\min g_n(y) = (-1)^{m_0+1} \mu_n(y) [\hat{f}(y) - f_n^*], y \in D \setminus \{x_1, x_2, \dots, x_n\} \quad (33)$$

where, y is an unsampled point; f_n^* is the target value; m_0 is a constant whose value depends on the basis function used (1 for cubic and thin plate splines, 0 for linear and multi-quadratic and -1 for Gaussian); $\mu_n(y)$ is the coefficient of the new term $\phi(\|x - y\|_2)$ in the surrogate $\hat{f}_n(x)$ if an unsampled point y is added. It is calculated as the n^{th} element of vector v , and v is calculated by solving the system of equations given by Eq. (34).

$$\begin{pmatrix} \phi_y & P_y \\ P_y^T & 0 \end{pmatrix} v = \begin{pmatrix} 0_n \\ 1 \\ 0_{d+1} \end{pmatrix} \quad (34)$$

$$\phi_y = \begin{pmatrix} \phi & \phi_y \\ \phi_y^T & 0 \end{pmatrix}; P_y = \begin{pmatrix} P & \\ y^T & 1 \end{pmatrix}; (\phi_y)_i = \phi(\|y - x_i\|_2); i = 1, \dots, n.$$

Minimizing the bumpiness function emphasizes exploration as well as exploitation depending on f_n^* . A large negative value of f_n^* makes the search global and focuses on exploration whereas, a value close to current optimal solution makes the search local and focuses on exploitation. Evaluation of bumpiness function is computationally expensive because obtaining μ_n by solving the system given by Eq. (34) is an $O(n^3)$ operation. However, the cost of this step can be improved by exploiting the structure of ϕ after which the operation becomes $O(n^2)$ (Björkman and Holmström, 2000).

5.3. Other approaches

The problem of adaptive sampling can be formulated as a DFO problem with the objective function being the difference between the true function and the surrogate (Cozad et al., 2014). The objective function is given in Eq. (35)

$$\max \left(\frac{f(x) - \hat{f}(x)}{f(x)} \right)^2, x_L \leq x \leq x_U \quad (35)$$

where, $\hat{f}(x)$ is the surrogate, $f(x)$ is the true function, and x_L and x_U are the bounds within which error is to be maximized.

Some approaches rely on ranking the exploration and exploitation and weighing both as per need. One such approach was recently proposed by (Garud et al., 2017). They propose a metric consisting of two separate measures for exploration and exploitation. For exploration, they use the sum of squares of the distance a new sample from all the previous samples. For exploitation, the impact of new sample added near an already sampled location is quantified with the help of a departure function.

$$\Delta_j(x) = \hat{f}(x) - \hat{f}_j(x), j \in S \quad (36)$$

where, $\hat{f}(x)$ is the surrogate built using all points in the sampled set S , $\hat{f}_j(x)$ is the surrogate built using all points except point j . Estimating the prediction variance of surrogate model using a technique called jackknifing, (Eason and Cremaschi, 2014) propose an adaptive sampling strategy. They use this strategy with a surrogate model built using ANN and choose samples locations that have high prediction variance. Advantage of this type of adaptive sampling is that it is not specific to the choice of surrogate model. A study of space filling sequential design methods can be found in (Crombecq et al., 2011). They propose a set of sequential sampling methods that shows comparable performance with stationary or one-shot experimental design.

6. Surrogate model validation

Assessing the reliability of surrogate model is one of the major concerns because having an inaccurate surrogate model can lead to waste of resources and have a bad effect on optimization, prediction or feasibility analysis. Surrogate model validation is the process of assessing the reliability of the surrogate model. In addition to assessing accuracy, validation techniques can be used to select a surrogate model from a set of candidate models and to tune hyper-parameters (such as correlation model parameters in Kriging). For problems of lower dimensions, a visual comparison between predictions and true value is possible. However, the difficulty in having enough data for visual comparison and inability to visualize predictions for problems over two dimensions necessitates more sophisticated approaches. As surrogate models cannot be validated on the same data with which those were built, surrogate models are built with the help of only a part of the available data. The remaining part of the data is used for testing the accuracy. The data set on which the model is built is referred to as training set and the set on which the model is tested is referred to as test set. The metrics used for quantifying the error on test set are referred to as validation metrics.

One possible approach for tackling this is using resampling strategies such as cross validation and bootstrapping. In cross-validation, available data is divided into k blocks containing an equal number of data points. Data from $(k-1)$ blocks are used as training set and data from the remaining block are used as a test set. The process is repeated for all possible combinations of $(k-1)$ blocks. Finally, an appropriate metric to quantify the error on test set such as the sum of squared errors is evaluated based on the accuracy of the model on test data that can act as an indicator of model adequacy. However, with limited data available, using part of the data for building the surrogate is not always possible. One such approach known as leave one out cross validation was used by (Donald R Jones et al., 1998). In this approach, the number of subsets k equals the number of data points or observations, thus leaving only one data point each time a surrogate is built. A sampling set is considered inadequate to build a quality surrogate if removal of one data point significantly affects the new model. A similar approach, but with allowing repeated samples in the training set is known as bootstrapping. By allowing repeated samples in the set used to build models, one can have a training set of the size

Table 4
Commonly used surrogate validation metrics.

Validation metric	Formula
Explained variance score	$1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)}$
Mean absolute error	$\frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} y_i - \hat{y}_i $
Mean squared error	$\frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} (y_i - \hat{y}_i)^2$
Median absolute error	$\text{median}(y_1 - \hat{y}_1 , \dots, y_n - \hat{y}_n)$
R ² score	$1 - \frac{\sum_{i=0}^{n_{\text{samples}}-1} y_i - \hat{y}_i^2}{\sum_{i=0}^{n_{\text{samples}}-1} y_i - \bar{y}^2}$
Relative average absolute error	$\frac{\sum_{i=1}^{n_{\text{samples}}} y_i - \hat{y}_i }{n_{\text{samples}} * \text{STD}}$
Relative maximum absolute error	$\frac{\max(y_1 - \hat{y}_1 , y_2 - \hat{y}_2 , \dots, y_n - \hat{y}_n)}{\text{STD}}$

equal to the size of actual data. Usually, number of subsets k chosen for bootstrapping is much higher than that for cross validation. Details on resampling methods for validation of surrogates can be found in (Bischl et al., 2012).

Validation metrics that are commonly used to quantify the error using the above-mentioned resampling strategies are the explained variance score, the mean absolute error, the mean squared error, the median absolute error, the R^2 score, the relative absolute error, and the relative maximum absolute error. These metrics with their respective mathematical equations are shown in Table 4 where y , $\hat{y}$, n_{samples} , and $\bar{y}$ denote true value, surrogate predicted value, number of samples and mean predicted value, respectively. Relative maximum absolute error indicates error in one part of feasible space. However, it is not a good indicator of the overall performance. Explained variance score equals R^2 score if mean of prediction error is zero. (Keresting et al., 2007) use normalized mean squared error as well as average Negative Log estimated Predictive Density (NLPD) for heteroscedastic Gaussian process regression that penalizes over-confident as well as underconfident predictions. In the same area, (Boukouvalas and Cornford, 2009) use Mahalanobis error that utilizes full predictive covariance avoiding the assumption of uncorrelated errors. (Yin et al., 2011) use mean absolute error as well as maximum absolute error for validation. For the case of multiple surrogates, (Felipe A C Viana et al., 2009) used prediction sum of squares (PRESS) as an estimator of root mean square error (RMSE) to pick the best surrogate. Their computational results reveal that PRESS becomes more and more useful for identifying the best surrogate as the number of sample points increases. PRESS vector $\tilde{e}$ is the vector of errors obtained from carrying leave one out cross validation. RMSE is predicted using Eq. (37).

$$\text{PRESS}_{\text{RMS}} = \sqrt{\frac{1}{n_{\text{samples}}} \tilde{e}^T \tilde{e}} \quad (37)$$

7. Software implementations

Several software implementations exist for DFO. List of these software can be found in (Rios and Sahinidis, 2013). Relatively less number of software implementations exist for surrogate model building for prediction and feasibility analysis. Few of the software implementations are listed followed by a recent DFO framework:

1) Automated learning of algebraic models for optimization (ALAMO) (Cozad et al., 2014)

ALAMO is a regression and classification methodology that builds simple and accurate surrogates from a minimal set of data points. It makes use of integer programming based subset tech-

nique to choose from a large number of possible input variables and their transformations. Starting from an initial stationary sample, ALAMO adaptively improves surrogate model by error maximization (discussed in Section 5.3). DFO is used for finding the location at which the error between the original model and the surrogate is maximum. Finally, ALAMO allows the user to make use of a priori knowledge about the model response and utilizes it to refine the model. This is achieved using constrained regression where a constraint is placed on the response of the regression model. This kind of problem is usually nonconvex in nature and therefore, a deterministic global optimization solver BARON (Tawarmalani and Sahinidis, 2005) is used to solve this sub problem. In addition to the ability of providing surrogate models from static data-sets, ALAMO also implements best subset selection methodology (discussed in Section 2.1.1). Details about constrained regression in ALAMO can be found in (Cozad et al., 2015). Detailed description of the ALAMO approach in the context machine learning approaches is given in (Wilson and Sahinidis, 2017).

2) Eureqa (Schmidt and Lipson, 2009)

Commercially available software Eureqa starts with an initial dataset and follows by carrying out a symbolic regression where the search is not restricted just to the coefficients but also to the form of regression model. It starts with a set of candidate regression models. Accuracy of these models is assessed by carrying out symbolic differentiation of models and comparing derivatives with the underlying data set. Based on the discrepancy, these models are then iteratively updated till a convergence criterion is met. Search through candidate regression models is carried out using algorithms similar to genetic programming details of which can be found in (Riolo et al., 2010) and (Riolo et al., 2011).

3) Surrogates Toolbox (Viana, 2010)

This is a general-purpose toolbox for surrogate modeling that consists of a number of third party software. This toolbox has four major capabilities. For designing experiments, it supports full factorial and LHS designs. For surrogate prediction, it supports surrogate models of type Kriging, Gaussian process for machine learning (GPML), radial basis neural network, and linear Shepard model. For surrogate model validation, it allows classical error validation including coefficient of determination, RMSE as well as cross validation. For optimization, it allows sensitivity analysis, contour estimation, and variants of EGO algorithm. List of third party software used for the above-mentioned purposes is as follows:

- Gaussian processes for machine learning (GPML) (Rasmussen and Williams, 2006a)
- Kriging (Lophaven et al., 2002a,b)
- SVR (Gunn, 1998)
- Linear Shepard model (Thacker et al., 2010)

4) Surrogate Modeling (SUMO) Toolbox for surrogate modeling and adaptive sampling

SUMO is a toolbox for surrogate modeling and adaptive sampling proposed by (Gorissen et al., 2010). The user is allowed to choose from surrogates such as Kriging, splines, support vector machines, and ANN. For hyper-parameter optimization, the toolbox supports particle swarm optimization, efficient global optimization, simulated annealing, genetic algorithm etc. The toolbox has model selection criteria such as cross validation, Akaike information criteria (AIC), leave-one-out cross validation etc. Additionally, it supports adaptive sample selection also known as active learning. Support for LHS and Box-Behnken sampling designs is provided.

5) Scikit-learn (Pedregosa et al., 2011)

scikit-learn is an open source library that implements a range of algorithms for machine learning, preprocessing, cross validation and visualization. For surrogate models, scikit-learn provides multiple regression models such as linear and support vector regression

Table 5
Relevant packages in R.

Package name	URL	Description
leaps	https://CRAN.R-project.org/package=leaps	subset selection using leaps and bound algorithm (Furnival and Wilson, 1974)
SIS	https://CRAN.R-project.org/package=SIS	sure independence screening (discussed in Section 2.1.1) for high dimensional problems
stats	https://stat.ethz.ch/R-manual/R-devel/library/stats/html/stats-package.html	linear and nonlinear regression model building and prediction, goodness-of-fit measures, model validation metrics
e1071	https://CRAN.R-project.org/package=e1071	support vectors regression and classification
neuralnet	https://CRAN.R-project.org/package=neuralnet	building surrogates and prediction using neural networks
subselect	https://CRAN.R-project.org/package=subselect	variable selection (Cadima et al., 2012)
lars	https://CRAN.R-project.org/package=lars	least angles regression, forward stagewise regression, and lasso
performanceEstimation	https://CRAN.R-project.org/package=performanceEstimation	Validation metrics
lhs	https://CRAN.R-project.org/package=lhs	Latin hypercube sampling
randtoolbox	https://CRAN.R-project.org/package=randtoolbox	Sobol and halton sequence
mlr (Bischl et al., 2016)	https://CRAN.R-project.org/package=mlr	Machine learning in R, regression and classification

and interpolation models such as Kriging and cross validation metrics to test the performance of surrogates.

6) Statistics and machine learning toolbox MATLAB

This toolbox supports linear and nonlinear regression, subset selection using goodness-of-fit measures and regularization, and support vector machines. It also includes validation metrics for testing performance of surrogates. In addition to these, this toolbox provides classification algorithms and non-parametric regression algorithms.

7) Software packages in R

R is a well-known open source platform for machine learning and data analysis and offers number of packages to choose from. Few of the packages are mentioned in Table 5.

8) ARGONAUT (Boukouvala and Floudas, 2016)

ARGONAUT addresses DFO problems where there is a partial or total lack of analytical expressions or closed form for objective function as well as constraints. Several steps of this iterative framework include bound tightening, variable selection, sampling (initial stationary sampling followed by adaptive sampling at locations suggested by the framework), surrogate model selection, and global optimization of the system to obtain new candidate points. The deterministic global optimization solver ANTIGONE is used for bounds tightening as well as global optimization. After bounds tightening, variable selection is done using sure independence screening (SIS) (discussed in Section 2.1.1). Important part of this framework is surrogate model selection where surrogates are chosen from a set of regression models such as linear, quadratic and polynomial models that are easy to optimize and interpolation models such as Kriging and RBF that are better at predicting data accurately.

8. Computational results and discussion

The purpose of computational experiments in this section is to illustrate the effect of following choices on the performance of surrogates in terms of their predictive ability.

1. For Kriging models, the choice of regression and correlation terms
2. Initial sample size
3. Initial sample design scheme

8.1. Test problems

In this comparison, the performance of Kriging and RBF surrogates is tested on problems of dimensions ranging from 2 to 30 over a box bounded domain of interest. Distribution of the problem sizes is shown in Table 6. The problems are from princetonlib (Princeton

Table 6

Problem size range and number of test problems.

Problem size	Number of problems
1–2	5
3–8	27
9–30	15

Table 7

Initial sample sizes and their denoted name.

Initial sample size	Size name
10 x problem dimensions	Size 1
20 x problem dimensions	Size 2

Library (2017)) and globallib (GLOBAL Library (2017)). The C version of the problems was obtained from <http://archimedes.chem.cmu.edu/?q=dfocomp>. All the problems chosen are nonconvex and smooth. Problem data such as problem size and variable bounds are obtained from the same web site.

8.2. Test setup and basis for comparison

For comparing the performance of RBF and Kriging surrogates, three different sampling schemes (Latin Hypercube Sampling (LHS), Sobol sampling, and Halton sampling) are used to obtain initial design. Of these three sampling schemes, LHS is a random sampling method whereas Sobol and Halton are quasi-random sampling methods. Therefore, 10 different random starting designs were obtained for LHS. Initial designs used were kept constant across all problems having the same number of dimensions to maintain consistency in comparison. To study the effect of initial sample size on the quality of surrogate, two different initial sample sizes (shown in Table 7) were used.

For two-dimensional problems, a grid of 2500 samples was used as a test set for prediction for RBF and Kriging surrogates. For problems with more than two dimensions, an LHS design with 1000 samples was used as a test set for prediction. For 10 random LHS designs, the median is considered as a representative of the performance. For analyzing the test results, the error factor (EF) is used as a measure of relative model performance.

$$EF = \frac{RMSE}{RMSE_{best}}$$

where, RMSE is the root mean squared error on test set and $RMSE_{best}$ is the minimum RMSE showed by the surrogate models under comparison for a particular problem. This error factor is then used to obtain Dolan–Moré plots (Dolan and Moré, 2002) or performance profiles of surrogates.

Table 8
Kriging surrogate models and their abbreviation.

Kriging trend	Correlation	Abbreviation
constant	squared exponential	KR11
constant	absolute exponential	KR12
constant	linear	KR13
linear	squared exponential	KR21
linear	absolute exponential	KR22
linear	linear	KR23
quadratic	squared exponential	KR31
quadratic	absolute exponential	KR32
quadratic	linear	KR33

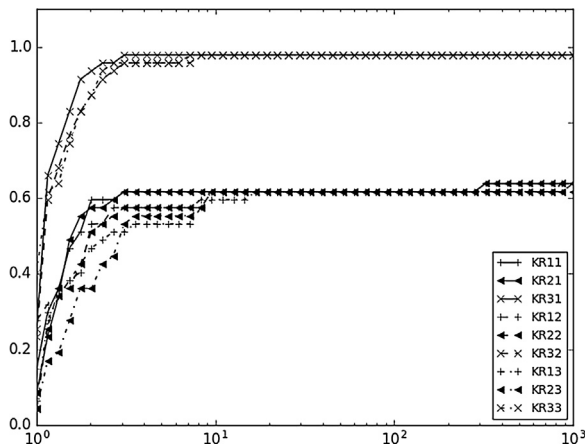


Fig. 1. Fraction of problems solved vs error factor for LHS design of size 1.

8.3. Surrogate model settings

For Kriging surrogates, three different types of correlation models (absolute exponential ($p_j = 1$ in exponential correlation model), linear, and squared exponential) and three different models for mean prediction term (constant, linear, and quadratic) were used that are shown in Table 3. Description of Kriging surrogate models and their abbreviations used is provided in Table 8. In addition to the choice of correlation model and mean prediction term, Kriging model needs information for hyper-parameter optimization. This information includes bounds on hyper-parameters, the initial guess for optimization of hyper-parameters and an appropriate optimization algorithm for minimizing the ML. In this comparison, bounds on all hyper-parameters were $[0.1, 10]$ and an initial guess of 1.1. DFO algorithm COBYLA (Powell, 1994) is used for optimizing ML function. The Kriging model is used from the toolbox scikit-learn (Pedregosa et al., 2011).

For the case of RBF, cubic basis function with a linear tail is used. To build RBF model, the system of equations given by Eq. (19) is solved.

8.4. Computational results

8.4.1. Choice of kriging regression term

Figs. 1–3 present Dolan–Moré performance profile of Kriging surrogates on test problems in terms of prediction accuracy. Value on vertical axis for each point in this plot represents the fraction of problems for which surrogates have EF less than the corresponding value on horizontal axis. Best case scenario for a surrogate would be represented by a vertical line at EF of 1. This would indicate that a particular surrogate model outperformed all other surrogate models for all test problems. For some of the problems, a surrogate may have an RMSE close to 0. To avoid numerical difficulties in evaluating EF, values of RMSE less than 0.000001 are approximated as 0.000001.

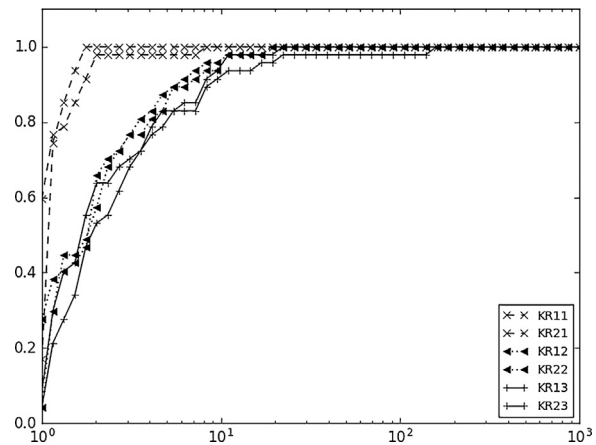


Fig. 2. Fraction of problems solved vs error factor for LHS design of size 1.

Fig. 1 compares all Kriging surrogates starting with a sample size of size 1 obtained using LHS design. KR31, KR32, and KR33 show a superior performance compared to other models. For a significant fraction of problems, all the surrogates having constant and linear regression terms have an EF of greater than 1000. This indicates that having quadratic regression term helps in improving prediction accuracy of Kriging models. However, including quadratic regression term needs $\frac{(d+1)(d+2)}{2}$ minimum sample points as it requires building of a fully quadratic model. This number gets large as problem size (d) increases. For example, for the problems with size 30, initial sample size of $10d$ is insufficient to build the model. This factor along with added computational expense with complex regression terms (discussed in Section 2.4.3) makes simpler regression terms an attractive choice. Another observation is that there is a little difference between KR31, KR32, and KR33 which indicates that as regression terms become complex, effect of correlation terms decreases. Because of these reasons, Kriging surrogates with quadratic regression terms are omitted in subsequent illustrations.

8.4.2. Choice of kriging correlation model

Fig. 2 compares Kriging models with constant and linear regression terms starting from samples generated from LHS with a size of size 1 and highlights the effect of correlation models on simpler regression terms. It can be observed that the performance of KR11, KR12, and KR13 is comparable with that of KR21, KR22, and KR23 respectively. This indicates that prediction accuracy using constant regression terms is comparable to that using linear regression terms. Second observation is that KR11 and KR21 show a superior performance compared to other Kriging variants. This indicates that squared exponential correlation terms show a superior performance compared to linear and absolute exponential correlation models.

8.4.3. Effect of initial sample size

In the next comparison, KR11, KR21 and RBF surrogate models are compared as the sample size is changed from size 1 to size 2 for an LHS design. Fig. 3 shows that, irrespective of the surrogate, increasing initial sample size improves prediction accuracy. This is intuitive because surrogate models are built utilizing more information.

8.4.4. Effect of sample design

Finally, it is important to illustrate surrogate prediction accuracy as initial sampling design is changed. Fig. 4 shows that, for initial sample of size 'size 1', Halton sampling shows worst performance for KR11 surrogate. Fig. 5 shows that performance of surrogates starting from three different sampling schemes is similar. This indi-

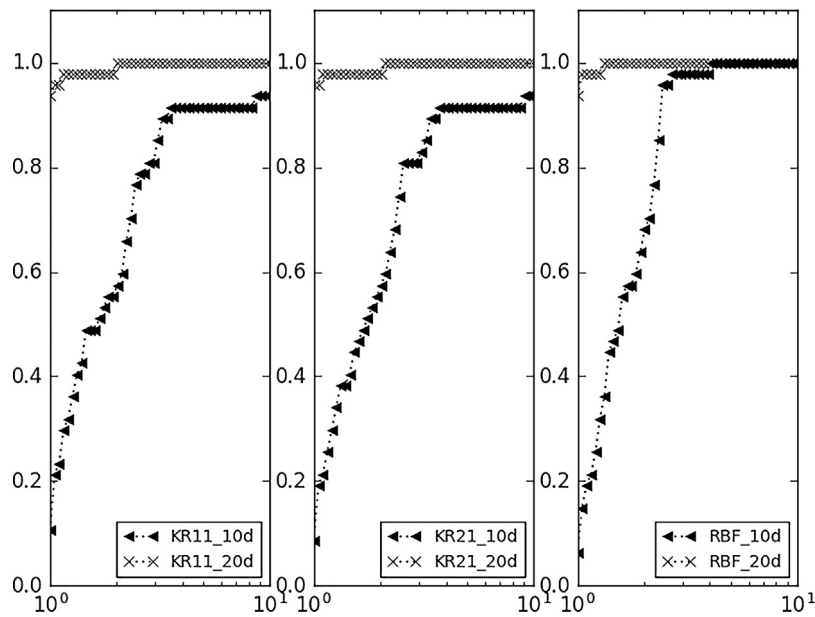


Fig. 3. Fraction of problems solved vs error factor for different sampling sizes.

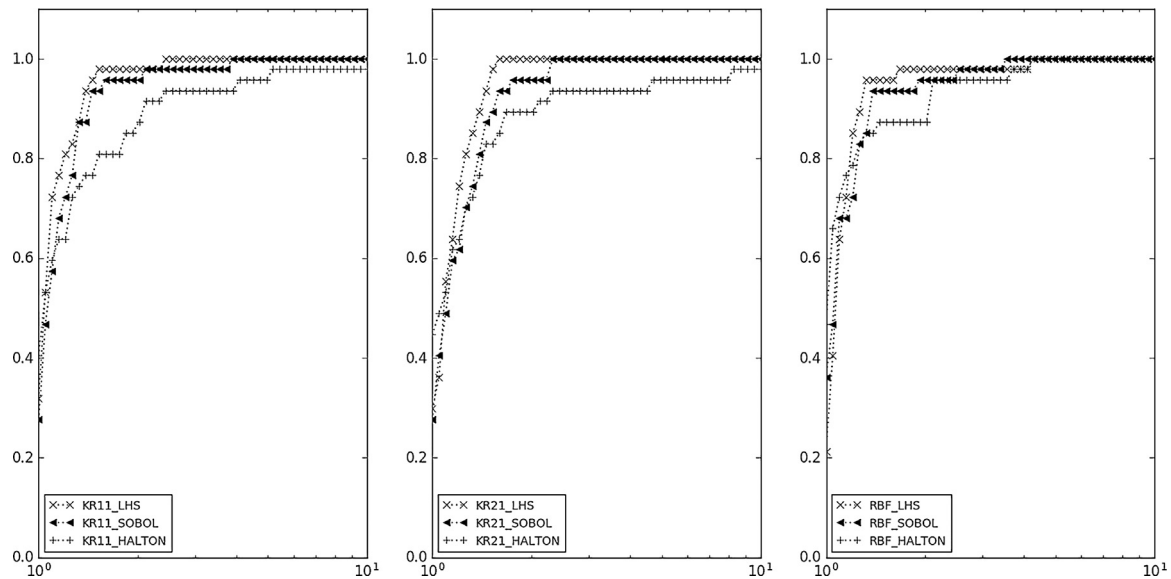


Fig. 4. Fraction of problems solved vs error factor for different sampling designs starting with size of 'size 1'.

cates that as sample size increases, sensitivity to sampling design scheme decreases.

Summarizing the discussion, a number of conclusions can be derived regarding the regression and correlation terms in Kriging surrogates, initial sample sizes, and initial sample design schemes. For correlation models, Kriging surrogates with squared exponential correlation model show superior performance. For the regression term, for an initial sample size of size 1, Kriging surrogates with quadratic regression term outperform the Kriging models with constant and linear regression terms. However, between the other two models, a model with constant regression term performs equally well as that with linear regression term. Kriging models with quadratic regression terms require determination of a full quadratic model. Therefore, these models have an increasingly high sampling requirement as problem size increases. Increasing initial sample size helps in better prediction accuracy irrespective of the surrogate model. Finally, for an initial sample size of size 1, surrogates built with initial samples obtained from LHS

and Sobol sampling schemes perform better than those obtained from Halton sampling scheme. However, as sample size increases, the prediction accuracy becomes less dependent on initial sample design scheme.

9. Summary

Surrogate models have attracted interest from multiple engineering and scientific disciplines and have found a wide range of application domains. However, the choice of surrogate model for the problem at hand is not straight-forward because of trade-offs associated with each surrogate. This choice is better understood if the problem at hand is classified as a prediction, optimization or a feasibility analysis problem based on the use of surrogate model. The difference of approaches for each of these classes from surrogate modeling point of view is emphasized. Relevant recent advances in each of the three classes are discussed with a focus on the use surrogate models.

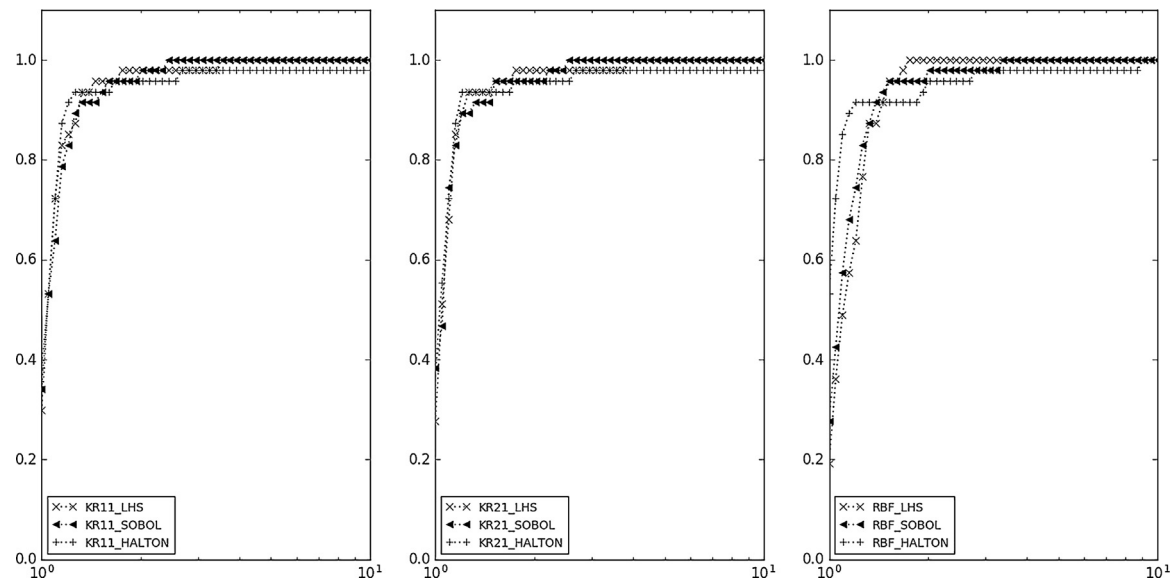


Fig. 5. Fraction of problems solved vs error factor for different sampling designs starting with size of 'size 2'.

For prediction using surrogates, mathematical formulation and recent advances in aspects such as prediction accuracy, numerical stability, and other computational aspects are discussed. Regression approaches such as linear and support vector regression are discussed followed by interpolation approaches such as RBF and Kriging. In the context of linear regression, a commonly used idea for preventing overfitting is subset selection. Several subset selection methodologies and ideas behind them are reviewed. In the context of blind Kriging, where there is no assumed form for regression term, Bayesian variable selection technique is introduced. For derivative-free optimization, model-based local and global search strategies are reviewed. Approaches, where surrogates are used within a framework, are discussed in addition to approaches that make direct use of surrogates. Literature displays successful application of surrogates inside trust-region like framework or a framework that exploits a certain quantitative measure such as EI function or bumpiness function. The idea of using EI function or bumpiness function has been shown to be successfully applied to address feasibility analysis problems. The quality of surrogates, as well as total computational cost of surrogate prediction, depends strongly on the sampling quality and efficiency. Adaptive sampling algorithms improve sampling cost by sequentially choosing the sample location starting from a stationary design. Details of adaptive sampling algorithms and use of surrogates for adaptive sampling is discussed. An important part of surrogate model building is assessing the quality of surrogate model at hand. Several fitness metrics are discussed. There are number of software implementations that provide the user with an easy access to ideas discussed in this review. These software are reviewed with a brief description and respective capabilities.

Finally, because of their wide applicability for all three classes of problems discussed in the paper, a computational study was carried out on Kriging and RBF surrogates. The results obtained agree with the common observation that having more samples improves the accuracy of surrogates. Additionally, Kriging surrogates with squared exponential correlation models were found to perform better than those with linear and absolute exponential correlation models. Kriging surrogates with quadratic regression model perform better than corresponding constant and linear regression models. However, Kriging surrogates with a constant trend term show a comparable performance with those with linear regression models.

Acknowledgement

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Appendix A.

Problem	Problem size	Convexity	Smoothness	Collection
3pk	30	smooth	nonconvex	princetonlib
allinit	3	smooth	nonconvex	princetonlib
arglinb	10	smooth	nonconvex	princetonlib
arglinc	8	smooth	nonconvex	princetonlib
biggs6	6	smooth	nonconvex	princetonlib
branin	2	smooth	nonconvex	princetonlib
brownbs	2	smooth	nonconvex	princetonlib
camel6	2	smooth	nonconvex	princetonlib
dixon3dq	10	smooth	nonconvex	princetonlib
extrosnb	10	smooth	nonconvex	princetonlib
genhumps	5	smooth	nonconvex	princetonlib
griewank	2	smooth	nonconvex	princetonlib
hart6	6	smooth	nonconvex	princetonlib
heart6ls	6	smooth	nonconvex	princetonlib
heart8ls	8	smooth	nonconvex	princetonlib
hilberta	10	smooth	nonconvex	princetonlib
hs038	4	smooth	nonconvex	princetonlib
hs045	5	smooth	nonconvex	princetonlib
hs110	10	smooth	nonconvex	princetonlib
least	3	smooth	nonconvex	globallib
nonmsqrt	9	smooth	nonconvex	princetonlib
oslbqp	8	smooth	nonconvex	princetonlib
palmer1c	8	smooth	nonconvex	princetonlib
palmer1d	7	smooth	nonconvex	princetonlib
palmer2a	6	smooth	nonconvex	princetonlib
palmer2c	8	smooth	nonconvex	princetonlib
palmer2e	8	smooth	nonconvex	princetonlib
palmer3c	8	smooth	nonconvex	princetonlib
palmer3e	8	smooth	nonconvex	princetonlib
palmer5b	9	smooth	nonconvex	princetonlib
qudlin	12	smooth	nonconvex	princetonlib
s211	2	smooth	nonconvex	princetonlib
s271	6	smooth	nonconvex	princetonlib
s272	6	smooth	nonconvex	princetonlib
s273	6	smooth	nonconvex	princetonlib
s276	6	smooth	nonconvex	princetonlib
s281	10	smooth	nonconvex	princetonlib
s282	10	smooth	nonconvex	princetonlib
s283	10	smooth	nonconvex	princetonlib
s291	10	smooth	nonconvex	princetonlib

s294	6	smooth	nonconvex	princetonlib
s295	10	smooth	nonconvex	princetonlib
s368	8	smooth	nonconvex	princetonlib
s370	6	smooth	nonconvex	princetonlib
s371	9	smooth	nonconvex	princetonlib
st_bs3	6	smooth	nonconvex	globallib
st_cqpjk2	3	smooth	nonconvex	globallib

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